

FFGFT in Brief

Foundations, results and status after the calculation review

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Contents

FFGFT in Brief	3
1 What FFGFT is	1
Ratios in natural units	1
Architecture in four layers	1
Resonance as the guiding idea: Euler's Tonnetz	1
Guidelines	2
What the corpus no longer claims	2
2 The basic form	3
Natural units, $\alpha = 1$ and the charge	3
Time-mass duality	3
The parameter ξ	4
Scales as ratios	4
Time structure and recursion	5
Geometry, Hilbert space, spectral theory	5
Field theory: Lagrangian density, mode operator, Schrödinger limit	5
Galois structure	6
3 Particles: the ratios	7
Lepton ladder	7
Galois identities, Koide and the four routes	7
Icosahedron and golden ratio: the φ skeleton	7
Neutrinos	8
Anomalous magnetic moments	9
Electroweak sector, Higgs, top	9
Quarks	9
Charge, hypercharge, generations, colour, photon	10
4 The translation into SI	10
One anchor	10
Electroweak scale	10
Fractal correction	11
E_0 and the SI value of α	11
Gravitational constant	11
5 Gravitation and cosmos	12
Black holes	12
Casimir and CMB	12
Cosmic sector: deliberately set aside	12
Galaxies	12
6 Quantum mechanics	13
7 Testable statements	14
8 What no longer holds	15
9 Open bridges	15
10 Status balance	16

11	Method and bookkeeping	16
12	Reading guide	17
13	Conclusion	18
14	Use of AI tools	18

FFGFT in Brief

Abstract

The Fundamental Fractal Geometric Field Theory (FFGFT) is first and foremost a ratio-based theory. It works in natural units, $\hbar = c = 1$, with the convention $\alpha = 1$ and a correspondingly redefined charge $e = \sqrt{4\pi}$. In this basic form there is exactly one dimensionless parameter, $\xi = 4/30000 = 1/7500$, and no fractal correction. Particles, couplings and gravitation appear as properties of a compact four-dimensional carrier $T^4/\mathbb{Z}_3$; the starting point is the time-mass duality $\tilde{T} \cdot m = 1$. The guiding idea is resonance: particles are oscillation modes, mass ratios are ratios of periods, $1/\xi = 2^2 \cdot 3 \cdot 5^4$ is a point in Euler's Tonnetz, and the fractal correction is the comma of the embedding into the continuum. The core statements are ratios: m_μ/m_e and m_τ/m_μ follow from ξ to within 0.5 and 1 %, the electroweak scale relates to the Planck energy as $\xi^4/(5\pi)$, the minimal length and minimal time relate to the Planck quantities as ξ , the neutrino mass differences agree to 1 %, and charge quantisation follows algebraically from the Galois structure of $\text{GF}(27)^*$. The golden ratio enters through the icosahedron: the weights of the fivefold rotation stand exactly as φ^8 and $2\varphi^4$, and $m_\tau/m_e = (74 + \frac{1}{45})\varphi^8$ agrees to 3×10^{-5} , within the measurement uncertainty. Only the translation into SI needs a measured value as anchor, the factor $K_{\text{frak}} = 74/75$ and conversion quantities such as C_{conv} ; this is also where the absolute values of α and G arise. This document summarises the corpus (Docs. 001–389, A series) as of 1 October 2026, after the complete recalculation. For every result it gives the formula, the numerical value, the comparison value, the status and the documents that carry it, and it states just as clearly what is set by choice, what remains open and what no longer holds. The cosmic sector (H_0 , Λ) is deliberately set aside.

Status markers

- [B]** Algebraically proved (transformation or proof without additional assumption).
- [K]** Numerically confirmed (verification script, comparison with measurements).
- [S]** Choice, open or speculative: motivated but not derived.
- [X]** Refuted or excluded. After a negated sentence the marker applies to the negated claim, not to the sentence.
- [Q]** Convention or adopted quantity.

The corrected state is authoritative: the correction register Doc. 190 up to R150 and release v1.4.4, in which every corrected passage carries a dated note with the previous value. Comparison values: CODATA 2022, PDG 2024. All numbers in this document are recomputed by the verification script `pruef_384_zusammenfassung.py`.

1 What FFGFT is

Ratios in natural units

The core of FFGFT is ratios. In natural units, mass, energy, temperature and inverse time are one quantity in four guises (A085), and every physical statement can be written as a dimensionless ratio: mass ratios, scale ratios to the Planck quantities, mixing angles, charge ratios. Such ratios do not depend on any choice of units and carry the content of the theory. Absolute values in SI are a second layer: they need a measured anchor and conversion quantities, and corrections appear there that do not occur in the ratios (A080, A130, Doc. 122). This summary follows that order: first the basic form and the ratios, then the translation into SI.

Architecture in four layers

FFGFT rests on three explicitly chosen basic assumptions (R56, Doc. 365): the time-mass duality $\tilde{T} \cdot m = 1$, the compact carrier T^4 and the identification with order 3, $\mathbb{Z}_3$. A fourth, motivated structural choice is the lattice D_4 ; Doc. 380 shows that among the comparison carriers tested with a $\mathbb{Z}_3$ action that has fixed points, D_4 has the lowest lattice energy **[K]**, but the choice remains a choice **[S]**. In addition there is exactly one dimensionless parameter, $\xi = 4/30000$. The fine-structure constant is set to $\alpha = 1$ by adjusting the charge unit; this is a convention, not a result **[Q]**. Galois fields, Hodge theory and representation theory serve as consistency checks, not as additional inputs (Doc. 365, layer 4). The spacetime dimension $3 + 1$ is an empirical input (P34).

Resonance as the guiding idea: Euler's Tonnetz

Behind all individual results stands one idea: whatever exists needs resonance. A particle is an oscillation mode on the compact carrier with integer winding numbers (resonance condition, Doc. 189); through $\tilde{T} \cdot m = 1$ every mass becomes a frequency and every mass ratio a ratio of periods. Ratios of periods are characters of cyclic groups, and the $\mathbb{Z}_3$ reduction turns them into arithmetic in $\text{GF}(3^k)$: harmonics and Galois classification are not analogous but two representations of the same object (Doc. 358, Theorem D **[B]**). What stands above is thus the geometry that possesses resonances; it fixes all ratios. The measured value is

the concert pitch: it says *where* the structure sounds, not *how* (Doc. 310; on the anchor, Section 4).

The model is Euler's Tonnetz (1739), the lattice of just intonation with the primes 2, 3, 5 as axes. It carries the theory at three points:

- **ξ is a lattice point.** $1/\xi = 7500 = 2^2 \cdot 3 \cdot 5^4$ lies in the 5-limit space at the coordinates $(-2, -1, -4)$ (Doc. 316 [K]). Seven of the nine Yukawa prefactors are 5-limit ratios like the intervals of just intonation (Doc. 060); the two exceptions, the primes 13 and 7, are in the Galois reading the harmonic primes of depth $k = 3$ and $k = 6$, and all nine lie in the grid $\{2, 3, 5, 7, 11, 13\}$ (Doc. 358, Theorem B [B]).
- **The mass ladder is a logarithmic resonance scale.** Because $\xi \ll 1$, the exponent axis is a musical scale: a step $\Delta p = 1/3$ is the fixed interval $\xi^{1/3} \approx 1/19.6$, the "third" of the particle spectrum (Doc. 189).
- **The comma is the embedding error.** Euler's spiral of fifths and thirds never closes, because $2^x 3^y 5^z = 1$ holds only for $x = y = z = 0$; commas remain [Q]. In a finite field, by contrast, every circle of fifths closes exactly, and a comma is impossible there (Doc. 358, Theorem A [B]). The musical comma and the fractal correction are therefore not a property of the Galois structure but of the embedding into the continuum; this is why K_{frak} appears only in the SI translation. The ξ cycle itself is defined rolled up, with the rational step $1/75$, and closes after exactly 75 turns (Doc. 315 [B]); the rotation number $74/75$ is the same value as K_{frak} .

The same structure recurs in coupling: undercritical, critical and overcritical coupling of two circuits correspond exactly to the 2×2 mixing of two states (Doc. 328 [B]); consonance is coupling geometry. The limits are recorded just as clearly: the harmonic reading is a representation, not an additional input; a Tonnetz with a finite limit covers only finitely many primes (Doc. 316), and hit rates of harmonic assignments with wide tolerance windows are a coverage effect, not evidence (Doc. 343).

Guidelines

Five rules determine how results are to be read:

1. **Ratios are what counts.** An absolute value in SI tests the conversion, not the structure (A130).
2. **FFGFT works with ξ alone.** Measured values from SI and the Standard Model serve only for conversion and comparison, and for that exactly one measured value is needed as anchor (Doc. 383).
3. **Integer structural choices are choices.** $\mathbb{Z}_3$, D_4 , the winding numbers or the exponent 10 in the cosmic sector are motivated choices, not calibrations.
4. **The cosmic sector is set aside.** H_0 and Λ are deliberately excluded from the derivation of constants (P39): the Standard Model does not derive them either, and none of the frameworks compared determines the ratio of Planck length to Hubble length from first principles.
5. **Explanation before precision.** What FFGFT yields are explanations: why masses stand in fixed ratios, why charge is quantised, why there are three generations, why Koide's $Q = 2/3$ holds, why the Born rule is a geometric quotient. The numbers test these explanations; they do not replace the calculations of the Standard Model.

What the corpus no longer claims

The calculation review has withdrawn several earlier claims:

- FFGFT is not a precision extension of the Standard Model. Where the Standard Model calculates, FFGFT gives the same values or uses the measured ones as anchors; a measurable ξ correction that would be more precise does not exist (R142, Doc. 382). Where it has free parameters, FFGFT supplies structure at the per-cent to per-mille level, not at measurement precision.
- FFGFT is not “postulate-free”: $\tilde{T} \cdot m = 1$, T^4 , $\mathbb{Z}_3$, D_4 and ξ are chosen.
- Precision tables of older documents in which prefactors had been fitted to measured values are not evidence (R138).
- The agreement of the SI value of α is a property of the chosen anchor, not an independent prediction (R135).
- The precision of G to 0.003 % is not evidence in its own right (R141).
- A ξ effect in Bell experiments or quantum gates that is measurable with present-day hardware does not exist (R142, R143).

The older documents are corrected accordingly; release v1.4.4 records the earlier wording.

2 The basic form

Natural units, $\alpha = 1$ and the charge

With $\hbar = c = \varepsilon_0 = 1$ (Heaviside–Lorentz) what remains of the fine-structure constant is the relation $e^2 = 4\pi\alpha$ [B]. The basic form sets $\alpha = 1$ [Q] and hence $e = \sqrt{4\pi}$: the charge is measured by the full surface of the sphere (A267). α does not disappear in the process; it moves into the charge unit,

$$\alpha = \left(\frac{e_{\text{hist}}}{e_{\text{geo}}} \right)^2 = \frac{r_e}{\lambda_C} \quad [\text{B}], \quad (1)$$

the ratio of the Coulomb and Compton spheres of the electron (Doc. 386). Geometrically, $\alpha = 1$ means that both spheres coincide, and with them the Bohr radius. In every calculation α stays on record through $e^2 = 4\pi\alpha$, one factor $\sqrt{\alpha}$ per vertex (A267). The bridge between α and ξ runs through the energy scale $E_0 = \sqrt{m_e m_\mu}$, the geometric mean of the two lightest charged leptons; in the basic form it reads

$$\xi E_0^2 = 1, \quad E_0^2 = \frac{1}{\xi} = 7500 \quad [\text{B}]. \quad (2)$$

This fixes the T0 energy unit and leaves no room for a correction factor.

Time-mass duality

In natural units the intrinsic time scale and the mass of a state are reciprocal:

$$\tilde{T} \cdot m = 1, \quad \tilde{T} = \frac{\hbar}{mc^2} \text{ (SI)}. \quad (3)$$

The relation describes every state as a snapshot, not as a law of motion (Doc. 365). A position-dependent mass in a gravitational potential is equivalent to a position-dependent time; “mass constant, space curved” and “space flat, mass variable” are two representations of the same situation (Doc. 078 [K]). For clocks and redshift the clock rate suffices; light also requires the fractal path lengthening, which carries the spatial half (Docs. 308, 389). These are representations, not ontological statements; mixed forms are admissible as long as clock rate and path together give $\gamma = 1$. On the carrier the duality holds mode by mode

exactly as a position relation $\lambda_4 \cdot m = 2\pi$ (Doc. 314 [K]). Reciprocity gives $E \rightarrow \infty \Leftrightarrow T \rightarrow 0$; since time is bounded from below by $T_0 = \xi t_P$, the horizon is an extremum and not a singularity (Doc. 306).

The parameter ξ

$$\xi = \frac{4}{30000} = \frac{1}{7500} = 1.3 \times 10^{-4}. \quad (4)$$

ξ is the only dimensionless parameter. It has several consistent readings, but none produces the number without an input:

- **geometrically** as the number 4 of lattice directions of D_4 over $3 \cdot 10^4$ (Docs. 004, 320);
- **as a Galois number** via

$$\frac{(r_\mu/r_e)^2}{\xi} = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)| = 8^2 \cdot 25 \cdot 27 = 43200 \quad (5)$$

(Docs. 336, 338), which is arithmetically exact [B] but depends on the chosen prefactors r_i ;

- **spectrally** as the smallest eigenvalue of the D_4 sub-operator, $\xi = \lambda_{\min}(\hat{F}_{D_4})$ (Docs. 322, 327), which is well defined provided the D_4 sector lies in the $\mathbb{Z}_3$ -invariant sector [B], but whose value is inserted, not derived.

Overall, ξ is a motivated choice [S]: in the words of Doc. 365, not completely free, but not yet completely derived either.

Check via the Higgs geometry

The Higgs geometry serves as a check, and it too is a pure ratio: with the self-coupling $\lambda_h = m_h^2/(2v^2) = 0.129$ one has $\xi_{\text{EFT}} = \lambda_h^2 v^2 / (16\pi^3 m_h^2) = m_h^2 / (64\pi^3 v^2) = 1.30 \times 10^{-4}$; the expression depends only on m_h/v and lies about 2.3 % below $4/30000$ (PDG 2024; 2.2–2.6 % depending on rounding; Docs. 354, A190, 385) [K]; the deviation is interpreted structurally [S] (R59). This is a consistency check to within a few per cent, not an exact derivation. The route goes back to the original vacuum formula of 2025 with ε_0 ; written out it is $\xi_{\text{EFT}} \cdot \alpha$, and with $\alpha = 1$ the same expression (Doc. 385) [B].

Scales as ratios

All scales of the theory stand in fixed ratios to one another. The minimal length, minimal time and maximal energy relate to the Planck quantities as

$$\frac{L_0}{\ell_P} = \frac{T_0}{t_P} = \xi, \quad \frac{E_{\max}}{E_P} = \frac{1}{\xi} = 7500. \quad (6)$$

The exponent 1 is derived [B] (R90): L_0 is the Compton range of the time-field mediator with mass $m_T = M_{\text{basis}}/\xi$; the choice $M_{\text{basis}} = m_P$ in the fundamental sector is open [S]. L_0 makes the number of levels of the fractal operator finite and hence $\hat{F}$ bounded [B] (Doc. 327). The electroweak scale relates to the Planck energy, and the lepton masses via the ladder (Section 3), likewise in pure ξ ratios (Docs. 149, 383):

$$\frac{v}{E_P} = \frac{\xi^4}{5\pi} = 2.012 \times 10^{-17} \text{ } (-0.23 \%), \quad \frac{m_i}{E_P} = \frac{r_i}{5\pi} \xi^{p_i+4} \text{ [B].} \quad (7)$$

With the measured E_P the lepton masses come out as 0.504, 104.8 and 1780.9 MeV ($-1.32, -0.80, +0.22\%$) [K]. The factor 10 in $v = E_P/(f^4 \cdot \frac{\pi}{2} \cdot 10)$ with $f = 1/\xi$ is fitted to the measured value. If the bare $v = 248.93$ GeV of the ladder is inserted instead of the measured v , the formula requires $10 K_{\text{frak}} \approx \pi^2$ instead of 10; the factor 10 would then be a mix-up of the bare ladder value with the SI comparison value [S] (Doc. 387).

On notation: here f is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units f is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{T0}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{T0} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, f cannot be set to 1 with justification (Doc. A135). Cf. Doc. 190, R147.

Time structure and recursion

The time structure runs step by step:

$$\xi_{n+1} = \xi_n (1 - 100 \xi_n), \quad \xi_0 = \frac{1}{7500}, \quad \xi_n \approx \frac{1}{100(n+75)}, \quad D(k) = \sum_{n < k} 100 \xi_n \approx \ln\left(1 + \frac{k}{75}\right). \quad (8)$$

In the frozen case with constant defect $d = 1/75$ the winding closes after exactly 75 turns; the rotation number is $74/75 = 1 - 100\xi$ [B]. This is the geometric origin of the value that appears as K_{frak} in the translation into SI (Section 4). In the running case the time winding becomes a logarithmic spiral with self-similarity ratio e , a continuous scale invariance with a pole at $n = -75$ (Doc. 295 [B]/[K]). The increments $d_n = 100 \xi_n$ decrease monotonically from $d_1 = 1/75$; d_1 is the largest, not the smallest increment, and the lower time bound is $T_0 = \xi t_P$ (Docs. 306, 307). Doc. 381 shows the telescoping product $\prod (1 - 100\xi_k) = \xi_n/\xi_0$ exactly [B], a second order $1/\xi_n \approx 100(n+75) + 100 \ln((n+75)/75)$ that is 50 to about 1000 times more accurate [K], and the constant $\psi'(75) = 0.01342$ between mass and time defect [K]. The assignment to a one-loop β function depends on the step convention [S]. The number 100 itself is chosen, not derived (Doc. 133, A040).

Geometry, Hilbert space, spectral theory

The state space is $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$; the translation between T^4 and $\mathcal{H}$ is bijective and lossless (Docs. 230, 270, 330). The torus is flat: the curvature vanishes pointwise via the metric descent [B] (R95). The spectrum separates the lattices: $\mathbb{Z}^4$ begins at $|k|^2 = 1$ with degeneracy 8, D_4 at $|k|^2 = 2$ with 24 (Doc. 314 [K]).

The fundamental fractal operator $\hat{F} = \bigoplus_n S_n \circ L_n$ with $S_n = r_n U_n$ and $r_n \in (0, 1)$ is bounded [B]. The $\mathbb{Z}_3$ pairing makes it normal; it is self-adjoint on the $\mathbb{Z}_3$ -invariant sector P_0 [B], with deficiency indices $(0, 0)$ and exactly one realisation. On the full space only the real part $(\hat{F} + \hat{F}^\dagger)/2$ is self-adjoint; using it is a choice [S] (R145, Doc. 327). The particle masses are eigenvalues of the mass operator $\hat{M}$, not of $\hat{F}$ (Docs. 322, 330).

Field theory: Lagrangian density, mode operator, Schrödinger limit

The field theory (A075, A070, A142; Docs. 019, 129) starts from the free Lagrangian of a scalar mass field δm on the torus and a fractal correction:

$$\mathcal{L} = \frac{1}{2}(\partial \delta m)^2 + \varepsilon \xi \left[\frac{f(\vartheta)}{2} (\partial \delta m)^2 - k^{\mu\nu}(\vartheta) \partial_\mu \delta m \partial_\nu \delta m \right]. \quad (9)$$

The geometric functions $f(\vartheta)$ and $k^{\mu\nu}(\vartheta)$ are not fixed [S]. The duality $\tilde{T} \cdot m = 1$ does not follow from a variational principle, since $\mathcal{L}_0 = T \cdot m - 1$ contains no derivatives; it remains a setting (Doc. 129).

$$\int \frac{1}{2}(\partial \delta m)^2 d^4x = \frac{1}{2}\langle \psi | \Phi^2 | \psi \rangle, \quad \Phi = \sum_i m_i P_i \quad \text{[B] [K]} \quad (10)$$

This bridge formula (A075) connects the Lagrangian and the mode operator: masses are eigenvalues of Φ , not parameters in the Lagrangian. This is the structural difference from the Standard Model, in which every mass enters through its own Yukawa coupling, and it is an explanation, not a statement of precision.

In FFGFT the Schrödinger equation is not fundamental. With the polar decomposition $\Phi \rightarrow \sqrt{\rho} e^{i\theta}$, $\psi \propto \sqrt{\rho} e^{i\theta}$ satisfies in the non-relativistic limit

$$i\hbar \partial_t \psi = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi; \quad (11)$$

its imaginary and real parts are the continuity equation and the Hamilton–Jacobi equation with quantum potential. The transformation is the known Madelung decomposition [Q], recomputed in the verification script [K]; the reading is that the equation describes the low-energy limit of the mode structure, for $E \ll E_\xi$ (A070, A075). The Dirac equation remains through the Clifford algebra; the γ matrices are its representation, and spin appears as a phase winding (A075).

For gravitation all matter fields couple conformally to the time field, $g_{\mu\nu} \rightarrow \Omega^2 g_{\mu\nu}$ with $\Omega = \mathcal{T}_0/\mathcal{T}$; the Dirac equation acquires the connection $\Gamma_\mu = \mathcal{T}^{-1} \partial_\mu \mathcal{T} = -\partial_\mu m/m$ [K] and turns into the standard form for constant $\mathcal{T}$ (A142). The Maxwell field is conformally invariant in four dimensions; for light the conformal coupling alone contributes nothing, it supplies the clock rate, and the spatial half comes from the path lengthening (A142, Doc. 389). Three statements of A075 and A142 were clarified: the limit $\xi \rightarrow 0$ is shown only for the scalar field, not for the full SM Lagrangian [S]; the modified Schrödinger equation $i\mathcal{T} \partial_t \Psi + \dots = \hat{H} \Psi$ is inhomogeneous with a dimensionful $\mathcal{T}$, homogeneous with $\mathcal{T}/\mathcal{T}_0 = 1/\Omega$, and then gives the standard form for $\Omega = 1$. Its amplitude follows the local energy density, not a probability (Doc. 230), and its phase runs at the local clock rate; this is the gravitational redshift (Doc. 078). Gravitation is therefore already contained in the equation [B][K], without an additional field and without a graviton: it shows itself as an effect in the redshift, read as curvature of space, as a change of mass (Doc. 078) or as fractal path lengthening (Doc. 041); a ξ addition of 2.34×10^{-10} to a_e , computed with $\alpha = 1$, and likewise the value 1.7×10^{-12} with the physical coupling $\alpha/(2\pi)$ are excluded by measurement, by 8.6σ or 10σ depending on the determination of α (Doc. 067).

Galois structure

The finite fields $\text{GF}(3^n)$ are the algebraic language of the mode classification on $T^4/\mathbb{Z}_3$. One has $|\text{GF}(9)^*| = 8$, $|\text{GF}(27)^*| = 26 = 2 \cdot 13$, $|\text{GF}(81)^*| = 80 = 2^4 \cdot 5$; $\text{GF}(9)$ is not a subfield of $\text{GF}(27)$, both lie in the compositum $\text{GF}(729)$. The Frobenius $x \mapsto x^3$ splits $\text{GF}(27)^* \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ into two fixed points $\{+1, -1\}$ and eight three-element orbits; on $\mathbb{Z}_{13}$ four orbits arise:

$$O_1 = \{1, 3, 9\}, \quad O_2 = \{2, 5, 6\}, \quad O_3 = \{4, 10, 12\}, \quad O_4 = \{7, 8, 11\} \quad \text{[B]}. \quad (12)$$

The 5 is the only primitive prime divisor of $3^4 - 1$, not the smallest one (Docs. 336, 338, 341). With the IPI correspondent D. Matzke only the algebra is jointly confirmed (orders in $G(3)$ and $G(6)$, Doc. 341); the physical assignments are on the FFGFT side.

3 Particles: the ratios

Lepton ladder

The charged leptons follow the ladder

$$\frac{m_i}{v} = r_i \xi^{p_i}, \quad r = \left(\frac{4}{3}, \frac{16}{5}, \frac{25}{9}\right), \quad p = \left(\frac{3}{2}, 1, \frac{2}{3}\right), \quad p_g = \frac{3}{2} \left(\frac{2}{3}\right)^{g-1}, \quad (13)$$

with $r_g = n_{\phi,g}^2/n_{\theta,g}$ from the winding numbers (3, 2), (5, 4), (9, 5) (Docs. 006, 317, 338) and a geometric sequence of exponents with starting value $D/2$ and ratio $2/3$ (Docs. 158, 018). The mass ratios contain no scale:

$$\frac{m_\mu}{m_e} = \frac{12}{5} \xi^{-1/2} = 120\sqrt{3} = 207.85 (+0.52 \%), \quad \frac{m_\tau}{m_\mu} = \frac{125}{144} \xi^{-1/3} = 16.99 (+1.03 \%) \quad \text{[K]}. \quad (14)$$

Formula and ratio calculation are [B], the agreement [K]; the r_i , p_i and winding numbers are choices [S] (R138). The step $1/3$ holds only from μ to τ ; e lies on an orbit of its own (Doc. 322). The ladder is written without K_{frak} ; in ratios it cancels exactly [B] (A040, A100, Docs. 070, 122).

Galois identities, Koide and the four routes

$(m_\mu/m_e)_{\text{ladder}}^2 = 43200$ is arithmetically exact; against the measured value a residual of -1.03% remains, an empirical discrepancy [K] whose cause is open [S] (R113). The Koide relation, itself a pure ratio,

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}, \quad (15)$$

holds for the measured masses to $Q - \frac{2}{3} = -0.017 \xi$ (PDG 2024; with $m_\tau = 1776.86 \text{ MeV} - 0.046 \xi$, Doc. 352) [K]; as a prediction of m_τ from m_e, m_μ it lies 0.4σ from PDG, undecided. From the ladder, Koide follows only approximately ($+8 \xi$).

In the $\mathbb{Z}_3$ circulant form $\sqrt{m_k} = c + d \cos(\theta + 2\pi k/3)$ one has $Q = \frac{2}{3} \Leftrightarrow d/c = \sqrt{2}$ [B]. The matrix-element route with $|A| = \sqrt{2}/3$ and $\theta = |A|^2 = 2/9$ (Doc. 367) hits $m_\tau/m_e = 3477.47$ to 3×10^{-5} [K] but gives $m_\mu/m_e = 206.7703$, which is 442σ from CODATA; the exact pair $(\sqrt{2}, 2/9)$ is excluded for m_μ/m_e [X] (Doc. 369, R121). The geometric derivation of $d/c = \sqrt{2}$ in Doc. 353 does not hold [S].

Doc. 369 compares four routes to the lepton masses (ladder, Koide matrix element, Galois, φ skeleton); routes 1 and 3 are the same calculation in two languages for m_μ/m_e . The following section shows the φ route.

Icosahedron and golden ratio: the φ skeleton

FFGFT is a $\mathbb{Z}_3$ theory from the ground up; the golden ratio enters through the icosahedron. Its rotation group A_5 is the smallest finite group in which a threefold and a fivefold rotation occur together without commuting (Doc. 370). The three acts as a lattice operation on the compact carrier, the five only as a phase in the unrolled space; A_5 is incompatible with the D_4 lattice ($5 \nmid 1152$), and the two meet in the field $\text{GF}(81)$ [B] (Docs. 364, 370).

The fivefold rotation R_5 mixes the three $\mathbb{Z}_3$ modes; the weights $p_j = |\langle v_j | R_5 v_e \rangle|^2$ are

$$p_0 = \frac{2}{9}, \quad p_1 = \frac{2+3\varphi}{9}, \quad p_2 = \frac{5-3\varphi}{9} \quad \text{[B]} \quad (16)$$

(Doc. 293, A120). The rational weight is the Koide phase, $p_0 = \theta = 2/9$; with $3\sqrt{p_0} = \sqrt{2}$ the same matrix element also gives the value of the Koide amplitude; that the amplitude is exactly $3\sqrt{p_0}$ is not derived [S] (Docs. 353, 367; R150). The two weights containing φ stand exactly in the ratios

$$\frac{p_1}{p_2} = \varphi^8, \quad \frac{p_0}{p_2} = 2\varphi^4, \quad 3\sqrt{p_j} \in \{\sqrt{2}, \varphi^2, \varphi^{-2}\} \quad \text{[B]} \quad (17)$$

(Doc. 368). These are pure numbers from the geometry, without a cosine and without a scale. The mass ratios carry this skeleton; the most precise hit is

$$\frac{m_\tau}{m_e} = \left(74 + \frac{1}{45}\right)\varphi^8 = 3477.469 \quad (\text{PDG } 3477.37 \pm 0.18; +3.0 \times 10^{-5}, 0.6\sigma) \quad \text{[K]}. \quad (18)$$

Ratio	Formula	Value	against PDG
m_τ/m_e	$74\varphi^8$	3476.42	$-2.7 \times 10^{-4} \ (-5.3\sigma)$
m_τ/m_e	$(74 + \frac{1}{45})\varphi^8$	3477.469	$+3.0 \times 10^{-5} \ (0.6\sigma)$
m_τ/m_e	$74\varphi^8(1 + \frac{27}{125}\xi)$	3477.468	$+2.9 \times 10^{-5} \ (0.6\sigma)$
m_τ/m_e	Koide, $\theta = 2/9$	3477.473	$+3.1 \times 10^{-5} \ (0.6\sigma)$
m_μ/m_e	$30\varphi^4$	205.62	-0.55%

On the φ route, m_τ/m_e is therefore as precise as on the Koide route, within the measurement uncertainty of m_τ , as a pure ratio without an anchor and without K_{frak} . The route is algebraic: since $\cos(2/9)$ is transcendental, no Koide ratio can be exactly a rational multiple of a power of φ [B]; the φ form approximates the same value algebraically, and the second layer ($\frac{1}{45}$ or $\frac{27}{125}\xi$) carries the remainder.

Limits of the φ route

The factors 74 and $\frac{1}{45}$ are observed, not derived from the $\mathbb{Z}_3/A_5$ framework [S] (Doc. 368). The 74 is the same number as in the numerator of the rotation number $74/75$ of the recursion (Doc. 381); its prime factor 37 demonstrably does not come from the icosahedron, because $\text{GF}(81)$ is not a subfield of $\text{GF}(3^{18})$ [B] (Doc. 370). For m_μ/m_e there is no φ route at this level: $30\varphi^4$ is off by 0.55 %, the ladder by 0.52 %, and the exact Koide pair is excluded there (above). The φ skeleton is not the earlier φ mass ladder with $m \propto \varphi^{\text{generation}}$ and $m_\mu/m_e \approx \varphi^{10}$, which is refuted [X] (Docs. 172, 281).

Neutrinos

Doc. 340 derives hierarchy and mixing from the four Frobenius orbits. The mass ratios are fixed,

$$m_1 : m_2 : m_3 = 1 : \sqrt{\frac{14}{3}} : 11, \quad \frac{\Delta m_{\text{atm}}^2}{\Delta m_{\text{sol}}^2} = \frac{120}{11/3} = 32.7 \ (-1.4\%) \quad \text{[K]}, \quad (19)$$

in normal hierarchy. Absolute values require a base mass; the assumed one is $m_\nu = \frac{\xi^2}{2}m_e = 4.54 \text{ meV}$ [S]. This gives $m_2 = 9.81$ and $m_3 = 49.96 \text{ meV}$, $\sum m = 64.3 \text{ meV}$, $\Delta m_{\text{atm}}^2 = 120 m_\nu^2 = 2.476 \times 10^{-3} \text{ eV}^2$ (-1.0% against 2.500×10^{-3} , Doc. 340) and $\Delta m_{\text{sol}}^2 = \frac{11}{3}m_\nu^2 = 7.565 \times 10^{-5} \text{ eV}^2$ ($+0.46\%$) [K]. The reactor angle gives $\sin^2 \theta_{13} = (25\xi)^{2/3} = 0.0223$ ($+1.4\%$) [K]. According to R108, ν_1 and ν_2 lie algebraically in the self-inverse Majorana layer and ν_3 is Dirac; this assignment of ν_3 to the orbit O_4 is in tension with Doc. 346, which assigns O_4 to the charged leptons, and Doc. 344 lists all neutrinos as Dirac [S]; that the algebraic self-inversion

physically means $\nu = \bar{\nu}$ is a further assumption [S]. For neutrinoless double beta decay this gives $m_{ee} \approx 6.0$ meV (phases zero), 0.13 meV in the destructive case, at most 14.4 meV, and exactly zero if all neutrinos are Dirac (Doc. 344). Older neutrino spectra (Docs. 006, 007) are excluded [X] (R109; Doc. 007). The A_5 eight-element spectrum gives $\sin^2 \theta_{23} \in \{4/9, 5/9\}$; $5/9 = 0.556$ lies close to the measured value for normal hierarchy, so maximal mixing $1/2$ is excluded (Doc. 372) [K]. θ_{12} is captured only in order of magnitude, the CP phase not at all [S]. The venting exponent $p = 5/24$ follows from the decomposition $\rho_4 \downarrow_{A_4} = \rho_3 \oplus \rho_1$ with $|A_5 : A_4| = 5$ [B] (Doc. 340).

Anomalous magnetic moments

The ratio line carries the result [K] (Docs. 018, 158): with winding exponents $5/3$ and $4/3$, $\Delta a(\tau-\mu)/\Delta a(\mu-e) = f^{1/3} - 1 = 18.57$, and anchored to the measured a_e, a_μ one obtains

$$a_\tau = a_e + \frac{144}{125} \frac{m_\tau}{m_\mu} (a_\mu - a_e) = 1.2811 \times 10^{-3}, \quad (20)$$

1.04×10^{-4} above the Standard Model value; present bounds are about 85 times wider, the decision is pending [S]. A fixed FFGFT addition $\Delta a_\mu = 251 \times 10^{-11}$ as the explanation of the former muon anomaly is superseded by the 2025 status; it would lie 3.4σ above today's difference [X] (Doc. 382).

Electroweak sector, Higgs, top

Here too the statements are ratios to the Z mass. On shell, $\sin^2 \theta_W = 1 - M_W^2/M_Z^2 = 2/9$ holds as an element of the A_5 eight-element spectrum (Doc. 372) [K]; this gives $M_W/M_Z = \sqrt{7/9}$, $M_W = 80.42$ GeV, 3.8σ from the 2024 world average, an explicitly stated, falsifiable tension. In the $\overline{\text{MS}}$ scheme the renormalisation running from $3/8$ gives $\sin^2 \theta_W(M_Z) = 0.2308$ (-0.19%) with two external inputs (Doc. 323) [K]. The ratios $m_h/M_Z = \frac{11}{8}$ ($m_h = 125.38$ GeV, $+0.15\%$) and $m_t/M_Z = (\frac{11}{8})^2$ ($m_t = 172.40$ GeV, -0.10%) and the trace rule $M_W^2 + M_Z^2 + m_h^2 = v^2/2$ ($+0.45\%$) are candidates [K]; the origin of the factors is a reading [S] (Doc. 373). The chain "symmetry breaking topologically enforced" is open at two points [S] (Doc. 355). Earlier Higgs formulas ($m_h = v\xi^{1/4}$, Doc. 041) are refuted [X] (R122).

Quarks

The quarks follow the same ladder form $m_q/v = r_q \xi^{p_q}$ with topologically motivated exponents [K], except for c and t , whose exponents are choices [S] (R138); the prefactors r_q are determined from measured values, their derivation is open [S] (R129, Doc. 373). The reproduction is therefore an encoding, not evidence; the table gives the values with $v = 246$ GeV:

Quark	r_q	p_q	Ladder ($v = 246$ GeV)	PDG 2024
u	6	$3/2$	2.27 MeV	2.16 MeV
d	$25/2$	$3/2$	4.73 MeV	4.70 MeV
s	$26/9$	1	94.8 MeV	93.5 MeV
c	2	$2/3$	1.284 GeV	1.273 GeV
b	$3/2$	$1/2$	4.26 GeV	4.18 GeV
t	$1/28$	$-1/3$	172.0 GeV	172.6 GeV

Light quark masses are scheme dependent; $\tilde{T} \cdot m = 1$ refers to pole masses. Λ_{QCD} does not follow from the ξ ladder [X] (R122): $v\xi^{1/3} = 12.6$ GeV, not 200 MeV. The CKM parameter

$\lambda = \xi^{1/6} = 0.2260$ lies 0.46 % from the measured value [K]; the exponent 1/6 is a motivated choice [S] (Doc. 336). The quark and hadron sector as a whole is an open bridge (Doc. 318, R76).

Charge, hypercharge, generations, colour, photon

Charge quantisation follows algebraically [B] (Doc. 346, R111): the QR orbits $O_1, O_3 \bmod 13$ are electrically neutral (neutrinos, antineutrinos), the NQR orbits O_4 and O_2 are charged (charged leptons, up quarks); together with $N \bmod 3$ this gives $Q \in \{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$, and "neutral and coloured" is excluded. The charge ratios are thus ratios of integers as well. The down quarks have no orbit of their own. Gell-Mann–Nishijima is not fully closed: $I_3 = \pm\frac{1}{2}$ follows [B], but the hypercharge of the charged leptons does not follow from the QR/NQR bit alone and is open [S] (R146, Doc. 347). The three elements of a Frobenius orbit give three generations [B] (Doc. 348); the earlier swap of generations 2 and 3 is an ordering artefact. $\mathfrak{su}(3)$ arises from the $\mathbb{Z}_3$ eigensectors [B]; $SU(3)$ is the smallest, not the only suitable group (Doc. 321). Photon and gluons are massless [B]; that masslessness means spin 1 does not follow from this and is open [S] (Doc. 339).

4 The translation into SI

One anchor

Because all scales stand in fixed ξ ratios (Section 2), a single measured value suffices for the translation into SI (Doc. 383). With m_e as anchor one obtains $E_P = 1.237 \times 10^{19}$ GeV (+1.34 %), $G = 6.50 \times 10^{-11}$ (−2.6 %), $\ell_P = 1.595 \times 10^{-35}$ m and $L_0 = \xi \ell_P$ [K]. The spread depending on the choice of anchor is the known residual of the ladder, not a second anchor:

Anchor	E_P	G	ℓ_P, L_0
m_e	+1.34 %	−2.62 %	−1.32 %
m_μ	+0.81 %	−1.60 %	−0.80 %
m_τ	−0.22 %	+0.45 %	+0.22 %
ν	+0.23 %	−0.46 %	−0.23 %

This fixes $L_0 = 2.155 \times 10^{-39}$ m, $T_0 = 7.19 \times 10^{-48}$ s and $E_{\max} = 9.16 \times 10^{22}$ GeV. With $\nu = 246$ GeV the absolute lepton masses lie at −1.18, −0.66 and +0.37 %.

Electroweak scale

In the mass calculation ν is an intermediate quantity. Depending on the route one obtains [K]:

Route	ν
Galois m_e from Doc. 338 (R104)	248.3 GeV
measured m_e	248.9 GeV
measured m_e and K_{frak}	245.6 GeV
$\xi^4 E_P / (5\pi)$	245.65 GeV

The comparison value $\nu = 246.22$ GeV is a Standard Model definition from G_F [Q] (R112); a convention-free comparison would only be possible via the muon lifetime τ_μ , which is open [S].

Fractal correction

$$K_{\text{frak}} = 1 - 100\xi = \frac{74}{75} = 0.98\bar{6}. \quad (21)$$

The factor appears only in the translation into SI: in ratios it cancels exactly **[B]** (A040, Docs. 070, 122), and in the basic form $\xi E_0^2 = 1$ there is no room for it. Its value comes from the geometry, as the rotation number of the frozen recursion (Section 2). It is confirmed **[K]** (R135): comparing the two routes to E_0 with the measured α requires $K = \xi m_e m_\mu \alpha^{-1} = 0.98650$, i.e. $n = 101.2$ instead of 100 in $1 - n\xi$. This factor is exactly the square of the bare reference energy in MeV (Section 4.4, Doc. 386, A130). The residual of 1.7×10^{-4} is not explained by measurement uncertainties and is open **[S]**. The *form* of the factor (additive $1 - 100\xi$ or multiplicative $(1 - \xi)^{100}$) is open again **[S]** (R139): the available witnesses do not separate the two forms more sharply than their difference of 8.8×10^{-5} . Equivalently, K_{frak} can be written in power form $(D/3)^{D/2} = 74/75$ with $D_f^{\text{eff}} = 2.97303$ **[K]** (Doc. 381); the spatial box dimension $D_f = 3 - \xi = 2.999867$ is a choice **[S]** that does not follow from the 100-step recursion (note in Doc. 322).

E_0 and the SI value of α

In SI, E_0 is the geometric mean of the two light lepton masses, bare and with the fractal correction (R72):

$$E_0^{\text{bare}} = \sqrt{m_e m_\mu} = 7.348 \text{ MeV}, \quad E_0 = \sqrt{\frac{m_e m_\mu}{K_{\text{frak}}}} = 7.397 \text{ MeV}. \quad (22)$$

With $\alpha_{\text{SI}} = \xi(E_0/1 \text{ MeV})^2$ one obtains without correction $\alpha^{-1} = 7500/E_0^2 = 138.9$, with correction 137.06, 1.7×10^{-4} from the measured value. The Galois form $\alpha^{-1} = 3700/27 = 137.037$ (+7.6 ppm, Doc. 338, R102) follows from $(E_0^{\text{bare}})^2 = m_e m_\mu \approx 54 \text{ MeV}^2 = |\text{GF}(3)^*| \cdot |\text{GF}(27)| \text{ MeV}^2$ together with $K_{\text{frak}} = 74/75$. Both agreements are properties of the anchor E_0 , not independent predictions **[K]** (R135).

The SI form requires the reference energy $E_0 \sqrt{\xi/\alpha} = 0.99992 \text{ MeV}$ (bare 0.99323 MeV, which is $\sqrt{K_{\text{frak}}}$ MeV). The MeV is not a geometric quantity but a unit of the SI chain **[Q]**: its numerical factor is the numerical value of the historical charge e , i.e. of the same unit in which α hides. If the correction is placed on this side, the side of the Coulomb sphere, the masses and the Compton sphere stay bare, and 1 MeV results to within 8×10^{-5} ; this assignment is a choice **[S]**, the residual remains open (Doc. 386). Third routes to E_0 and closed α formulas from ξ alone are withdrawn **[X]**.

Gravitational constant

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}, \quad C_{\text{conv}} = 7.783 \times 10^{-3}, \quad K_{\text{frak}} = 0.986. \quad (23)$$

G is not an independent constant: its form and order of magnitude follow from ξ and the one anchor **[K]** (Docs. 012, 180, 383). C_{conv} is the SI conversion of the one-anchor chain; its corpus value additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The agreement with CODATA to 0.003 % arises only with this residual and is not precision evidence in its own right (R141); it holds with the rounded corpus value $K_{\text{frak}} = 0.986$ of the G formula, with $74/75$ it would be +0.07 %. Explaining the residual structurally is open **[S]**. From $\tilde{T} \cdot m = 1$ one has $m_{\text{char}} \cdot G = \xi^2/4$ for the characteristic mass of the anchor (R58, A142); for arbitrary m the relation holds only if G runs along (Doc. 350). A mass singularity with finite G is excluded **[B]** (Doc. 350), while physical validity in the black-hole regime is open **[S]**.

5 Gravitation and cosmos

Black holes

Equating the Compton and Schwarzschild radii gives, purely geometrically,

$$M_{\text{coll}} = \frac{m_P}{\sqrt{2}}, \quad r_s = \sqrt{2} \ell_P, \quad S_{\text{BH}}(M_{\text{coll}}) = 2\pi \text{ nat} = \frac{2\pi}{\ln 2} \text{ bit} = 9.065 \text{ bit} \quad [\text{B}] \quad (24)$$

(Docs. 325, 329). S_{BH} equals Matzke's length in Planck units; Matzke's threshold is smaller by $\sqrt{2}$, $n_{\text{thr}} = 6.41$ (R144). As a universal bit number n_{thr} is refuted [X] (R88). The ratio of sector to thermal information is mass independent, $\ln 3 \approx 1.099$ [B]. The Hawking temperature corresponds to the bit energy $\hbar c/L$ at the horizon scale [K]; only modes with triality 0 can be emitted, and confinement and Hawking selection are one principle [B] (Doc. 325). Greybody factors and back-reaction are open.

Casimir and CMB

The Casimir energy density between ideal plates is $-\pi^2 \hbar c / (720 d^4)$; the form with 240 is the pressure. At the length L_ξ the ratio to the CMB density, $\pi^2 / (720 \xi) = 102.8$, is an identity, because L_ξ itself is determined from the CMB density and ξ ; it is not measurement evidence (R136). The QFT Casimir energy does not supply a factor $4/3$ [X] (Doc. 371). The temperature $k_B T_{\text{CMB}} = \frac{16}{9} \xi (1 - \frac{275}{4} \xi) \text{ eV}$ matches only in the unit eV and with an underived coefficient [S] (R137); the eV brings the electron mass along as a second anchor. In the basic form T_{CMB}/m_e depends on a single number. The candidate $T_{\text{CMB}}/m_e = (8\pi)^{1/4} \xi^{5/2}$ meets the FIRAS value to 0.12σ , together with the H_0 form gives $T_{\text{CMB}}^4 = 16 H_0 m_e^3$, and fixes L_ξ without the CMB; the prefactor is not derived [S] (Doc. 388). The factor 3 in front of the peak sequence $\{1, 6, 14, 26\}$ is a choice; the sequence itself is read off the data (R139).

Cosmic sector: deliberately set aside

The structural form $H_0/v_e = \pi^2 \xi^{10} = 1.75 \times 10^{-38}$ gives $H_0 = 66.8 \text{ km/s/Mpc}$; the exponent 10 is a motivated choice (P20), so the statement is conditional and not a prediction. Equally conditional are $\Lambda^* = 1/R_H^2$ and the recasting of the 10^{123} question as a structural question (Doc. 312). Excluded are the Λ formulas of older documents, $H_0 = \xi c$, $z = \xi x$ and $z \approx \xi \ln(d/\ell_P)$ [X] (R137; Doc. 156). The redshift $1+z = e^{\xi x}$ is achromatic [K], but the length in x is not specified from ξ ; the supernova time dilation is $1+z$ for all z [B] (R128). Supernovae, BAO and the CMB temperature are degenerate between expansion and intrinsic time unfolding; the decision is theoretical, not empirical (Doc. 267).

Galaxies

The acceleration scale $a_0 = c^2 \xi^{10} / (4 \bar{\lambda}_e) = 1.03 \times 10^{-10} \text{ m/s}^2$ lies 14 % below the empirical value and inherits the choice P20. The gas-dominated dwarf galaxy DDO 154 matches to 0.2 % [K]; for spiral galaxies deviations of about 19 % remain open. "Bullet Cluster passed" is not supported: with baryonic masses the peak lies at the gas (note in Doc. 308). Light deflection reproduces the GR value 1.75" [K]: clock rate and fractal path lengthening contribute 0.875" each, together they give $\gamma = 1$ (Doc. 308); the purely conformally flat metric of Doc. 143 would give 0 (Doc. 143). With the logarithmic form $\ln(m/m_0) = \Phi/c^2$ one also has $\beta = 1$, and the perihelion shift is that of GR [K]; the form itself is a setting [S] (Doc. 389).

6 Quantum mechanics

In quantum mechanics FFGFT is deterministic. The rules and numerical values of standard quantum mechanics remain; their interpretation changes at four points:

- **Measurement is a pole transition, not a collapse.** The interaction with the apparatus drives the state point to one of the poles; which one is fixed by the apparatus variable λ on the same axis, $A(z, \lambda) = \text{sgn}(z - \lambda)$. Every single measurement is thus determined, even when λ is unknown (Docs. 230, 175) [K].
- **The Born rule is a geometric quotient.** $(1 + z)/2 = |\alpha|^2$ is the share of the axis $[-1, 1]$ whose λ values lead to the pole $+1$ [K]; by Archimedes it is at the same time an area fraction of the Bloch sphere [B]. It appears as a frequency only in series of measurements with uncontrolled λ . Randomness is epistemic, not ontological (Docs. 365, 367, 370); its closest relative in the literature is Aerts' hidden-measurement model [Q].
- **Gates are geometric rotations.** The three Pauli matrices correspond to three rotations on the Bloch cylinder; CNOT is a geometric conditioning in which the rotation of one cylinder point depends on the z value of the other. Since $\{H, T, \text{CNOT}\}$ is universal, every gate can be reproduced deterministically and without repetition, up to the noise limit (Docs. 230, 175) [K].
- **Noise is disordered phase, and the limit applies to both evaluations.** A spectrum with fixed amplitudes and unresolved phases looks like white noise although every component is fixed, and with known phases it can be recomputed exactly [K]; vacuum noise can thus be a superposition of deterministic harmonics [S]. The process is deterministic and can be recomputed up to the noise limit: in the feedback the phase error grows with every step and reaches the limit after about $\pi/(2\epsilon)$ steps (Doc. 183) [S]. Beyond it, outcomes can no longer be distinguished. The deterministic reading needs no repeated measurements, since every single outcome is fixed. The probabilistic evaluation, however, gets no further: averaging over many runs runs into the same floor, whose natural scale is ξ (Doc. 343, Theorem G') [S].

Bell and Kochen–Specker exclude only local or non-contextual hidden variables [Q]. The FFGFT measurement is contextual, because the determination lies in the interaction, and for several qubits the correlation is a property of the closed geometry, not a local quantity in $\mathbb{R}^3$ (Doc. 230, R126). Operationally, FFGFT and standard quantum mechanics give the same predictions up to a ξ correction.

The Bell violation remains: the fractal damping reduces S only by an amount of order ξ , $\Delta S/S \approx 10^{-5}$ to 10^{-4} depending on the form, and establishes no local bound (R142). The only documented hardware measurement (IBM, 28 May 2026, Doc. 147, 2048 shots per run):

Device	Runs	$\langle S \rangle$	Standard error	Share of $2\sqrt{2}$
ibm_kingston	15	2.7396	0.0071	96.9 %
ibm_marrakesh	10	2.7016	0.0138	95.5 %
pooled	25	2.7244	0.0078	96.3 %

The Bell violation is confirmed [K]; the gap to $2\sqrt{2}$ (0.088 on ibm_kingston, 0.104 pooled) is decoherence, 160 to several thousand times larger than the ξ effect, and two identical chips differ by 0.038. The ξ effect cannot be resolved with present-day hardware [S].

CHSH values given earlier from alleged runs in 2025 and a ξ value fitted to them are unsupported and removed (R143). The deviation of the π gate by $100\xi = 1.33\%$ is absorbed by the pulse calibration of real qubits and is not directly measurable (R142).

Shor in the deterministic reading. Shor's algorithm splits into the preparation, a modular exponentiation with $O(n^3)$ gates and almost the entire cost (share $1 - 1/(n + 1)$, 99.95 % for

2048 bits [K]), and the quantum Fourier transform with $O(n^2)$ gates, a wave operation that extracts the period as a resonance (Doc. 176). In theory it factorises in $O((\log N)^3)$ [K]; a general exponential quantum advantage, however, is not supported (Doc. 176). The advantage lies solely in the period finding. Preparation and error correction cost more on quantum hardware than classically: the preparation by a fixed factor, the error correction by an overhead that, according to the threshold theorem, grows only slowly. The advantage would take effect only for numbers for which the hardware is lacking today (RSA-2048: 8.6×10^9 gates, 4.1×10^6 qubits [K]); in the reachable range a classical computer factorises just as fast. As an apparatus the quantum computer is an analogue computer with quantum-mechanical precision: a qubit is a two-level resonator, every gate a resonant drive, and error correction averages over syndrome rounds. This averaging runs into the same resolution floor as every evaluation; that it falls below it at scale is what the threshold theorem claims, but it has not been shown [S]. Whether a net advantage remains is thus decided by the error correction alone. If ξ is the floor, the limit of a resonance-based computer lies at $N_{\max} \approx 1/\xi = 7500$, about 13 bits, and the time advantage vanishes structurally [S] (Doc. 343, Theorem G'). There are alternatives for the computational operations themselves: the Fourier transform runs optically or in any resonator (Doc. 176), analogue matrix folding and resonator locking compute with resonances instead of gates (Doc. 186), as do resonance computers such as the BAW Ising machine (Doc. 362). They are subject to the same resolution floor. The calculations of standard quantum mechanics remain usable; FFGFT interprets them as resonance processes. What is withdrawn is only the earlier claim that a separate T0 or ξ factorisation method yields results distinguishable from Shor's (R65); ξ as a line width of phase estimation is refuted for the preregistered setup, the line remains open (P44).

7 Testable statements

Quantity	FFGFT	Measured / bound	Status
Ratios (basic form)			
m_μ/m_e	207.85	206.77 (+0.52 %)	[K], r_i [S]
m_τ/m_μ	16.99	16.82 (+1.03 %)	[K]
Koide Q	2/3	-0.017ξ	[K]
m_τ from Koide	1776.97 MeV	1776.93(9), 0.4σ	open (footnote 1)
m_τ/m_e (φ skeleton)	$(74 + \frac{1}{45})\varphi^8 = 3477.47$	3477.37(18), 0.6σ	φ^8 [B], factors [S]
v/E_p	$\xi^4/(5\pi)$	-0.23%	[K], factor 10 fitted
ξ from m_h/v	1.30×10^{-4}	$4/30000$ (-2.3%)	check [K]
$\Delta m_{\text{atm}}^2/\Delta m_{\text{sol}}^2$	32.7	33.2 (-1.4%)	[K]
$\sin^2 \theta_{13}$	0.0223	0.0220 ($+1.4 \%$)	[K]
$\sin^2 \theta_{23}$	5/9	normal hierarchy	[K]
a_τ	1.2811×10^{-3}	bound 85 times wider	open (footnote 2)
M_W/M_Z	$\sqrt{7/9}$, 80.42 GeV	3.8σ (world average)	tension
m_h/M_Z	11/8, 125.38 GeV	125.20 ($+0.15 \%$)	candidate [K]
m_t/M_Z	$(11/8)^2$, 172.40 GeV	172.57 (-0.10%)	candidate [K]
λ_{CKM}	$\xi^{1/6} = 0.2260$	0.2250 ($+0.46 \%$)	[K], exponent [S]
CHSH deviation	$\Delta S/S \sim 10^{-5}-10^{-4}$	hardware noise $\sim 10^{-2}$	not resolvable
π gate	1.33 %	in pulse calibration	not measurable
Absolute values (SI translation)			
Δm_{atm}^2	$2.476 \times 10^{-3} \text{ eV}^2$	2.500×10^{-3} (-1.0%)	[K]
Δm_{sol}^2	$7.565 \times 10^{-5} \text{ eV}^2$	$+0.46 \%$	[K]
m_{ee} ($0\nu\beta\beta$)	6.0 / 0.13 / 0 meV	nEXO 5–20 meV	[S], nEXO decides

Quantity	FFGFT	Measured / bound	Status
Σm_ν	64.3 meV	cosmological, model dependent	[K]
α^{-1}	137.06	137.036 ($+1.7 \times 10^{-4}$)	property of the anchor
τ_μ without $G_{F_I} \nu$	–	MuLan	open [S]

Footnote 1. Koide with the measured m_e and m_μ gives $m_\tau = 1776.97$ MeV, 0.4σ from the PDG mean $1776.93(9)$ MeV [K]. This does not decide the test: the two measurement methods, threshold scan (BESIII, 1776.91 MeV) and pseudo-mass (Belle II, 1777.09 MeV, with $m_{\nu_e} = 0$), scatter by 0.18 MeV, twice the stated uncertainty. Only when the methods agree does the value test Koide (R113, Docs. 351, 352).

Footnote 2. a_τ has not yet been measured; there is only an exclusion interval. The best bound is CMS 2024 with $-0.0042 < a_\tau < 0.0046$ (95 % CL), i.e. 0.0088 wide. The FFGFT value lies 1.04×10^{-4} above the Standard Model value 1.1772×10^{-3} ; the interval is about 85 times wider than this difference, and both values lie inside it [K]. The question can be decided only by a measurement about a hundred times more precise, for instance at Belle II (Doc. 382).

8 What no longer holds

The following statements of older documents are refuted or withdrawn [X]; the documents are corrected accordingly (Doc. 190, release v1.4.4).

Foundations: ξ as necessarily derived, a Casimir factor $4/3$, the number 100 as derived, a minimal time increment d_1 , self-adjointness of $\hat{F}$ on the full space, a field tower $\text{GF}(3) \subset \text{GF}(9) \subset \text{GF}(27)$, the exclusion of fivefold symmetry in four dimensions.

Particles: precision tables with fitted prefactors as evidence, the φ mass ladder with $m \propto \varphi^{\text{generation}}$ (not the φ skeleton), the exact pair $(\sqrt{2}, 2/9)$ for m_μ/m_e , Λ_{QCD} from the ξ ladder, earlier Higgs formulas, the Higgs route as a derivation of ξ , the W and Z masses as a prediction, fermion statistics from the D_4 geometry, the neutrino spectra of Docs. 006 and 007, a fixed muon addition $\Delta a_\mu = 251 \times 10^{-11}$, Gell-Mann–Nishijima as fully closed.

Gravitation and cosmos: the precision of G as evidence in its own right, n_{thr} as a universal number, the Casimir-CMB ratio as measurement evidence, the Λ formulas, $H_0 = \xi c$ and $z = \xi x$, “Bullet Cluster passed”.

Quantum mechanics: local realism or $S \leq 2 + \varepsilon$, a hardware confirmation of the ξ effect, the CHSH values 2.8275 and 2.8278, a factorisation method specific to FFGFT.

9 Open bridges

Doc. 190 keeps the open points in a list of bridges (as of R150). In summary:

- Basic form:** the origin of the number 100 and of the fitted factor 10 in ν/E_p (Docs. 149, 387); $M_{\text{basis}} = m_p$ (R90); self-adjointness of $\hat{F}$ on the full space (R145); an adversarial D_4 specificity test (R95); the spectral dimension $d_s = 1.86$ versus ≈ 2 .
- SI translation:** the residual 1.7×10^{-4} between the K_{frak} required by α and $1 - 100\xi$, and the form of the factor (R135, R139); the residual of 8×10^{-5} in the reference energy 1 MeV (Doc. 386); the residual of the one-anchor chain for G (R141).

3. **Particles:** τ_μ directly from FFGFT as the only convention-free test of the weak sector (R112); the cause of the decreasing lepton residuals (R113, R121); the Koide amplitude $d/c = \sqrt{2}$ from the geometry, equivalent to the equipartition of the mass sum between the symmetric mode and the non-trivial pair (Doc. 353, R150); the factors 74 and $\frac{1}{45}$ of the φ skeleton and the question whether the icosahedral phase and the Galois order have a common root (Docs. 368, 370); the hypercharge of the charged leptons (R146) and of the Higgs doublet (R148); the quark prefactors and the hadron sector (R76, R129); K_{had} (R123); two-loop running of α_s ; m_{Pl} and $\alpha_{\text{em}}(M_Z)$ from ξ ; spin 1 of the photon.
4. **Gravitation and cosmos:** greybody factors and back-reaction (R85); a physical Casimir signature of L_ξ (R136); T_{CMB} independent of units (R137); forward selection of the CMB peaks and the C_ℓ projection (P29–P31); the exponent 41/4 or 10 forward (P20); light elements without a hot early phase; Ω_{DM} model-neutrally.
5. **Method:** ξ as a line width (P44); clean-up of the H_0 language (P16, P34).

10 Status balance

Status	Results (selection)
[B]	$e^2 = 4\pi\alpha$ and $\alpha = r_e/\lambda_C$; $\xi E_0^2 = 1$ in the basic form (Doc. 386); charge quantisation from the Legendre symbol (Doc. 346); $SU(2)_L$, $U(1)_Y$ and $\sin^2\theta_W _{\text{GUT}} = 3/8$ (R80); $\mathfrak{su}(3)$ from $\mathbb{Z}_3$ sectors (Doc. 321); three generations as a Frobenius orbit (Doc. 348); self-adjointness of $\hat{F}$ on P_0 (R145); exponent in $L_0 = \xi \ell_p$ (R90); $m_i/E_p = \frac{\ell_i}{5\pi} \xi^{p_i+4}$ (Doc. 383); rotation number 74/75 and telescoping product of the recursion (Doc. 381); A_5 weights with $p_0 = \theta = 2/9$, $p_1/p_2 = \varphi^8$, $p_0/p_2 = 2\varphi^4$ (Docs. 293, 367, 368); $S_{\text{BH}}(M_{\text{coll}}) = 2\pi \text{ nat}$ and $\ln 3$ (R144); supernova time dilation $1+z$ (R128).
[K]	Lepton ratios and Koide; m_τ/m_e from the φ skeleton to 3×10^{-5} ; neutrino ratios, mass differences and θ_{13} ; a_τ ratios; v/E_p ; Higgs check of ξ to about 2.3 %; Higgs and top candidates; λ_{CKM} ; value $K_{\text{frak}} = 74/75$ and its appearance only in the SI translation; form and order of magnitude of G ; Bell violation on hardware.
[S]	$\tilde{T} \cdot m = 1$, T^4 , $\mathbb{Z}_3$, D_4 , ξ , r_i , p_i and winding numbers; the factors 74 and $\frac{1}{45}$ in the φ skeleton; the number 100 and the form of K_{frak} ; factor 10 in v/E_p ($10 K_{\text{frak}} \approx \pi^2$ with the bare v); $M_{\text{basis}} = m_{p_i}$; correction on the unit side (Doc. 386); base mass m_{v_i} ; quark prefactors; hypercharge of the charged leptons and of the Higgs doublet; spin 1 of the photon; exponent 10 in the cosmic sector; CMB factor 3.
[Q]	$\hbar = c = 1$ and $\alpha = 1$ with adjusted e ; $v = 246 \text{ GeV}$ as Standard Model definition; the anchor m_μ/m_e is QED dependent; Maxwell dynamics adopted; C_{conv} and the MeV as SI conversion.
[X]	see section "What no longer holds".

11 Method and bookkeeping

The corpus labels every statement with a status marker. Newer documents have a verification script with assertions, many of them null tests. The register Doc. 190 records corrections to older documents and open bridges; since R133 the source documents themselves are corrected; up to release v1.4.4 each passage carried a dated note with the previous value. Excluded predictions are not rewritten but annotated. From 29 September to 1 October 2026 the entire corpus was recalculated; in v1.4.4 about 235 documents carry a correction box and the A series carries notes in the text, while the cleaned documents

contain only the corrected text. The consequences for the classification are recorded in R135 to R146, R148 and R150. Guard rails:

- A comparison value is not an input.
- A form $X = \xi^N$ alone says nothing (P35).
- Predictions are preregistered with fixed degrees of freedom (P44).
- Every prediction names its anchor (R113).
- An absolute value in SI tests the conversion, a ratio tests the structure (A130).

12 Reading guide

Topic	Documents
Foundation	365 (four layers), 190 (register), 180/189 (chain $\xi \rightarrow L_0$), 383 (one anchor)
Units, α	A080/A085 (natural units), A130/A267 ($\alpha = 1$), 122 (ratio and absolute value), 386 (where α hides), 385 (Higgs–vacuum route), 387 (deviations)
Masses	006 (ladder), 336/338 (Galois), 351/352 (PDG, lepton residuals), 367/369 (matrix element, four routes)
Icosahedron, φ	291/292/293 (place and origin of $2/9$), 364 (A_5 and D_4), 368 (φ skeleton), 370 (why the icosahedron)
Electroweak	323 (Weinberg running), 372/373 ($2/9$, Higgs, Yukawa), 355 (symmetry breaking)
Charge, colour	321 ($SU(3)$), 346/347 (charge, Gell-Mann–Nishijima), 348 (generations)
Neutrinos, $g - 2$	340, 343/344 (Majorana/Dirac), 158/382 (a_τ , status 2025)
Gravitation	012/013 (G , SI), 325/329 (black holes), 350, 388 (CMB temperature in the basic form), 389 (parabolic flight and representations)
Structure	314 (lattices), 322/327/330 (spectral theory), 380 (D_4), 381 (recursion)
Field theory	019/129 (Lagrangian), A075 (bridge formula, Schrödinger and Dirac limits), A070 (low-energy limit), A142 (gravitation, time field)
Time	295 (log spiral), 306/307 (reciprocity, time in state space)
Quantum mechanics	230/231 (Hilbert space, measurement), 175 (qubit state spaces, deterministic readout), 183 (noise as phase), 023 (Bell), 147 (hardware), 176 (Shor: preparation and transformation), 343 (quantum computer as analogue computer)
Resonance, harmonics	060 (musical parallels), 189 (resonance condition), 310 (resonance geometry), 315 (Euler spiral), 316 (Tonnetz), 328 (coupling regimes), 358 (harmonics and Galois)
Delimitation	309 (scale anchor), 267 (degeneracy), 377 (interfaces), 378 (peer review)

13 Conclusion

FFGFT is a theory of ratios and yields explanations, not greater precision than the Standard Model. In natural units with $\alpha = 1$ it replaces the free parameters of the lepton sector by a structure with one parameter ξ and chosen integers, places lepton masses, the electroweak scale, the minimal length and the Planck scale in fixed ξ ratios, and obtains charge quantisation algebraically. Its guiding idea is resonance: particles are modes of the carrier, ξ is a point in Euler's Tonnetz, and the fractal correction is the comma of the embedding. The ratios agree to within a per cent down to per mille; through the icosahedron, m_τ/m_e carries the φ skeleton and agrees to 3×10^{-5} . The translation into SI needs one anchor, the fractal correction and conversion quantities; this is where the absolute values of α and G arise, and where a known residual lies. Open are the origin of the number 100, the origin of the fitted factor 10, the form of the fractal correction and its residual against the MeV, the hypercharge of the charged leptons, the hadron sector and everything that depends on the cosmic sector. Decisive tests in the coming years are m_{ee} (nEXO), a_τ (Belle II), the sum of neutrino masses and M_W . The corpus labels choices as choices and refutations as refutations; this summary reflects that state.

14 Use of AI tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author.

Verification script: 2/python/Dok384_Skripte/pruef_384_zusammenfassung.py — all numbers in this document recomputed.